Preventing and Responding to Weather-Related Injuries and Illnesses

From R. Rehberg, *Sport First Aid*, 6th ed. (Champaign, IL: Human Kinetics, 2026), 135-149.

Searing lightning, sweltering heat, and numbing cold are an accepted part of outdoor seasons in sports. Even so, coaches should not overlook the serious illnesses and injuries that can result from these conditions. Lightning, heatstroke, and hypothermia can be life-threatening, and frostbite can lead to disfiguration. Fortunately, almost all of these conditions can be prevented. As a coach, you play a key role in not only preventing but also quickly identifying and providing appropriate first aid care for temperature-related illnesses and lightning injuries.

Temperature-Related Illnesses

Before learning the sport first aid specifics for temperature-related illnesses, you need to first understand how body temperature is regulated. The body's temperature changes through several processes, including metabolism, convection, conduction, radiation, and evaporation (see figure 11.1). These methods of temperature regulation are described as follows:

- *Metabolism*—As the body's cells work and use energy (metabolism), heat is produced. When a person is active, the body temperature rises due to an increase in metabolic rate.
- *Convection*—In convection, heat is lost or gained by air (wind) circulating around the body. If the air temperature is warmer than the body, the body will gain heat; if the air temperature is cooler, the body will lose heat.
- *Conduction*—When the body touches warmer or colder objects, it gains or loses heat; this process is called conduction. For example, sitting in a warm whirlpool will cause body temperature to rise; sitting on a cold, metal bench will lower body temperature.
- *Radiation*—Radiation occurs when heat is gained through contact with electromagnetic waves, such as from the sun. The degree of cloudiness and the angle of the sun can influence the sun's radiative effects. Heat can be lost through radiation from the body to the environment whenever the environmental temperature is less than body temperature.
- *Evaporation*—Sweating is the body's built-in mechanism for cooling itself. However, it's only effective if the sweat actually evaporates from the skin. Humidity (the amount of moisture already in the air) directly influences how

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much sweat will evaporate. The more humid the environment, the less sweat will evaporate, and the more difficult it is for the body to lose heat. Equipment and type of clothing can also affect sweat evaporation. For example, a helmet will prevent heat from the head from being dissipated through radiation, evaporation, and convection.

With this understanding of body temperature regulation, the following sections explore what can happen when body temperature goes awry.

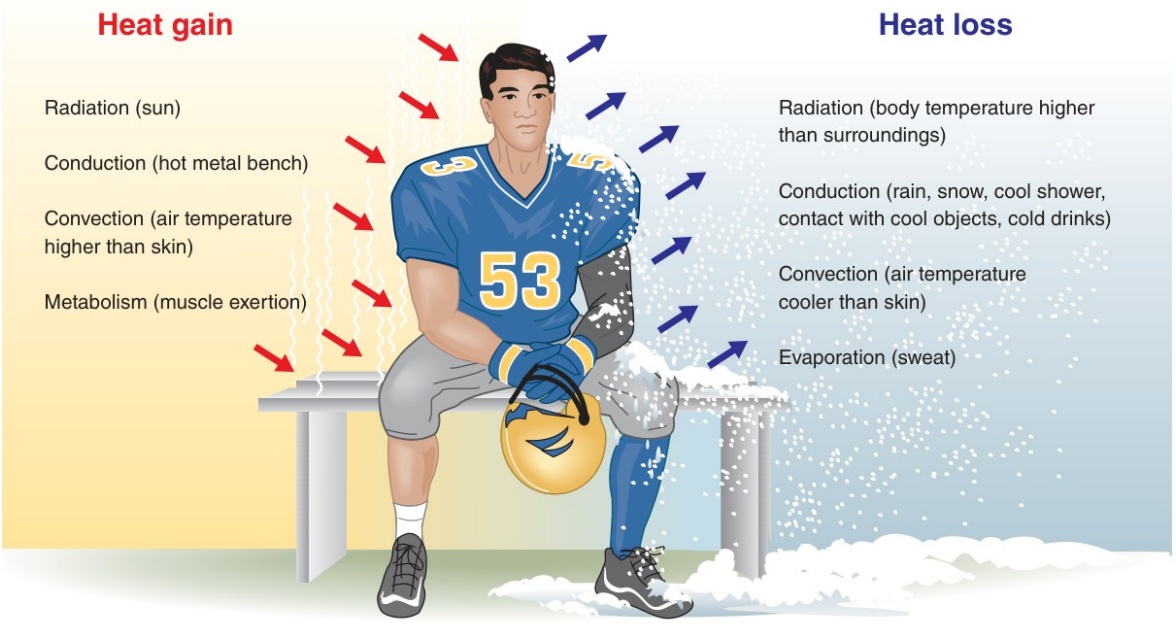


Figure 11.1 *Methods of body temperature change.*

Exertional Heat-Related Illnesses

Consider the following examples of heat-related illnesses:

After experiencing heat exhaustion and leaving practice early the day before, perhaps he felt he couldn't afford to take it easy during drills and conditioning, even though it was another day of stifling heat. So, he pushed himself through practice. It was hard; he vomited several times. Finally, it was time to head inside to the long-awaited air-conditioned environment. There, he developed symptoms of heatstroke, including weakness and rapid breathing. Although he was immediately evaluated and treated by athletic trainers and then transported to a hospital, Korey Stringer, offensive tackle for the Minnesota Vikings, was unresponsive at the time of arrival. His body temperature had soared to over 108 degrees Fahrenheit, resulting in multiorgan system failure and, finally, death.

In 1997, within the span of one month, three college wrestlers died trying to make weight. The February 20, 1998, issue of the *Morbidity and Mortality Weekly Report*, published by the Centers for Disease Control and Prevention (CDC), concluded that all

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three victims wore vapor-impermeable suits and exercised vigorously in hot environments. “These conditions promoted dehydration and heat-related illness,” stated the report, which noted that the body temperature of one of the wrestlers was 108 degrees Fahrenheit at the time of death.

Athletes can limit heat-related injuries and illnesses by taking proper precautions before participating in outdoor activity. Some of these precautions are outlined next.

Prevention of Exertional Heat-Related Illnesses

The Korey Stringer Institute recommends five pillars of exertional heatstroke prevention (NFHS and Korey Stringer Institute 2015). They include hydration, body cooling, work-to-rest ratios, acclimatization, and education. Key factors related to these pillars include the following:

Monitor Weather Conditions, and Adjust Practices Accordingly

Wet-bulb globe temperature (WBGT) is considered the gold standard of environmental monitoring. Multiple organizations, including the U.S. Department of Defense and the American College of Sports Medicine, recommend this method for environmental monitoring (Cooper et al. 2017). WBGT relies on humidity, radiant heat, ambient air temperature, and wind speed to display a more accurate temperature. Most U.S. high schools have developed plans for practice modifications according to the WBGT temperature. Typically, an athletic trainer (on staff or assigned to the school) monitors the WBGT temperature in real time and alerts the coach of practice or game modifications if needed.

Figure 11.2 shows the specific air temperature and humidity percentages that can be hazardous. This information can be used to help determine playing modifications in the absence of a WBGT device. Keep in mind that exertional heat-related deaths have occurred in football players at temperatures as low as 82 degrees Fahrenheit with a relative humidity index at only 40 percent. If heat and humidity are equal to or higher than these conditions, make sure athletes are acclimated to the weather and are wearing light practice clothing. Schedule practices for early morning and evening to avoid higher heat in the middle of the day. In addition, remember that information presented in figure 11.2 is less accurate than an onsite WBGT device, and it does not account for all factors that the device would use; however, it can still serve as a guideline for activity planning and modifications. Table 11.1 is an example of the NJSIAA (New Jersey State Interscholastic Athletic Association) heat participation policy. You can use this sample to create your own heat participation policy and help prevent exertional heat-related illness; it is important to first research your state’s policies and procedures.

In addition, having appropriate work-to-rest ratios (the amount of time spent involved in exercise versus the amount of time spent in recovery) should be modified as environmental conditions become extreme. Environmental extremes should be

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measured using WBGT, for an accurate measure of the heat stress that the athlete will be experiencing during exercise in the heat. Modifications of work-to-rest ratios in extreme environmental conditions include increasing the number of rest breaks, the duration of rest breaks, and having unlimited access to hydration.

Figure 11.2 *Heat index chart.*

Temperature (°F) across columns; relative humidity (%) down rows.

RH	80	82	84	86	88	90	92	94	96	98	100	102	104	106	108	110
40	80	81	83	85	88	91	94	97	101	105	109	114	119	124	130	136
45	80	82	84	87	89	93	96	100	104	109	114	119	124	130	137	
50	81	83	85	88	91	95	99	103	108	113	118	124	131	137		
55	81	84	86	89	93	97	101	106	112	117	124	130	137			
60	82	84	88	91	95	100	105	110	116	123	129	137				
65	82	85	89	93	98	103	108	114	121	128	136					
70	83	86	90	95	100	105	112	119	126	134						
75	84	88	92	97	103	109	116	124	132							
80	84	89	94	100	106	113	121	129								
85	85	90	96	102	110	117	126	135								
90	86	91	98	105	113	122	131									
95	86	93	100	108	117	127										
100	87	95	103	112	121	132										

Caution (yellow); Extreme caution (peach); Danger (orange); Extreme danger (red).

Source: National Weather Service. Available: https://www.weather.gov/ama/heatindex.

Table 11.1 *Sample Heat Participation Policy**

WBGT reading	Flag	Risk for heat illness	Activity and rest break guidelines
Under 80 °F	Green	Very low	Normal activities—Provide at least three separate rest breaks each hour (minimum duration: 3 min each).
80.0-85.0 °F	Yellow	Low	Use discretion for intense or prolonged exercise; watch at-risk players carefully; provide at least three separate rest breaks each hour (minimum duration: 4 min each).
85.1-88.0 °F	Orange	Moderate	Maximum practice time is 2 hours. For football, lacrosse, and field hockey: All helmets and shoulder pads must be removed for practice and conditioning activities. If the WBGT rises to this level during practice, football players may continue to work out wearing football pants without changing into shorts. For all sports: Provide at least four separate rest breaks each hour (minimum duration: 4 min each).
88.1-90.0 °F	Red	High	Maximum duration of practice is 1 hour. For football, lacrosse, and field hockey: No protective equipment may be worn during practice, and no conditioning activities are allowed. For all sports: No conditioning is allowed, and you must have 20 minutes of rest breaks distributed throughout the hour of practice.
>90.0 °F	Black	Very high	No outdoor workouts are allowed. Delay practice until a cooler WBGT level is reached.

*Please note that these WBGTs may vary depending on the location.

Adapted by permission from Guidelines, Policies and Procedures, 2022-2023, NJSIAA, accessed January 31, 2024, https://www.njsiaa.org/sites/ default/files/documents/2022-08/NJSIAA%20Policies%20and%20Procedures%20%2722-%2723%20-%20With%20accepted%20changes.pdf.

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Acclimate Athletes to Exercising in High Heat and Humidity

If you are located in a warm-weather climate or have practices during the summer, athletes need time (about 7-10 days) to adjust to high heat and humidity. During this time, hold short practices at low to moderate activity levels, and provide fluid and rest breaks every 15 to 20 minutes. The National Athletic Trainers' Association's Position Statement: Exertional Heat Illnesses (Casa et. al. 2015) offers more specific guidelines for acclimating high school athletes to hot environmental conditions; the recommendations are as follows:

- *Days 1 and 2*: Single 3-hour practice, or one 2-hour practice and one 1-hour field session with helmets only
- *Days 3 and 4*: Single 3-hour practice, or one 2-hour practice and one 1-hour field session with helmets and shoulder pads
- *Day 5*: Single 3-hour practice, or one 2-hour practice and one 1-hour field session with full pads
- *After day 5*: One day between days with multiple practices; less than 5 hours total practice time; and less than 2 hours walk through

The NATA specifies that any athlete who misses a practice for any reason must still complete acclimation guidelines as listed previously. Therefore, the athlete must pick up where they left off before missing a practice.

Switch to Light Clothing and Less Equipment

Athletes stay cooler if they wear shorts, white T-shirts, and less equipment (particularly helmets and pads). Equipment blocks the ability of sweat to evaporate. It's especially important for athletes to wear light clothing and minimal equipment while they are acclimating to the heat.

Prepare for Hot Practices

If practice must happen during times when heat may be an issue, plan accordingly for the conditions. Having access to cooling methods is essential for treating heat-related conditions; these methods include the following:

- *Cold-water immersion*—This method is the most effective way to cool an athlete in an emergency. It uses a tub, such as a small inflatable pool filled with water or any other vessel big enough to submerge an athlete up to their neck. Ideally, the tub should be filled with water before practice, and ice should be available to be added if needed. Water in the tub should be approximately 33.8 to 78.8 degrees Fahrenheit.
- *Tarp-assisted cooling oscillation*—This method, called the taco method, should be used when cold-water immersion is not available. In lieu of a rigid cold tub, many organizations recommend the taco method. The athlete lies on a tarp while volunteers hold the sides up. The tarp is then filled with ice water. The

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volunteers then use their knees to oscillate the water in the tarp to aid with cooling.

- *Cold shower*—This method is another way to cool an athlete in the event that cold-water immersion is not available. This method requires the athlete to remove unnecessary clothing, be positioned under a shower or hose, and get drenched in cold water.
- *Ice towels*—This method uses towels completely soaked in ice water; when an athlete needs cooling, the towel is placed over their shoulders. Although ice towels are easier to acquire and transport than a cold-water immersion tub, they are less effective at cooling.

Other cooling methods may include instant ice bags; ice packs placed on the inner thighs, armpits, and head; or ice vests. While these methods may help in an emergency, cold-water immersion should still be considered the primary choice for emergency cooling. In addition, you should ensure the drinking water provided to the athletes has ice in it; if individual bottles are used, keep them in a cooler.

Identify and Monitor Athletes Prone to Heat Illness

Athletes who are particularly prone to exertional heat illness include those who have previously experienced a heat illness and those with the sickle-cell trait. These athletes should be continuously monitored during activity. Dehydrated, overweight, heavily muscled, or deconditioned athletes, as well as athletes taking certain medications (antihistamines, decongestants, some asthma medications, certain supplements, and attention-deficit/hyperactivity disorder medications) are at risk. Closely monitor these athletes, and make sure they drink plenty of fluids. Rest dehydrated athletes until they have become rehydrated (see the following section on adequate hydration).

Strictly Enforce Adequate Hydration

Athletes can lose a great deal of water through sweat. If this fluid is not replaced, the body will have less water to cool itself and will become dehydrated. Dehydration not only increases athletes' risk for heat illness, but it also decreases their performance. In fact, athletic performance may worsen after only 2 percent of the body weight is lost through sweat. For example, dehydrated athletes may experience decreased muscle strength, increased fatigue, decreased mental function (e.g., concentration), and decreased endurance. Signs and symptoms of dehydration include the following:

- Thirst
- Flushed skin
- Fatigue
- Muscle cramps
- Apathy
- Dry lips and mouth

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- Dark urine (should be clear or light yellow)
- Feeling weak

Most athletes don't feel thirsty until they have lost 3 percent or more of their body weight in sweat (water). By that time, their performance will have started to decrease, and their risk of exertional heat illness will have increased. Therefore, instead of relying on athletes to drink enough fluids on their own, you should remind them to drink fluids at regular intervals. In addition, athletes may not drink enough fluid to replenish the water lost through sweat during activity in high heat and humidity. The most effective way to determine the amount of fluid lost through sweat is for athletes to weigh themselves with minimal clothing on before and after practices and competitions that take place in high heat and humidity. For every pound lost, an athlete should drink 16 to 24 ounces of water (DHS 2021). For additional aid in proper hydration, see your organization's licensed health care provider (athletic trainer, team physician). If hydration is a persistent issue, consult with the athlete's parent or guardian, or the appropriate health care provider.

Replenish Electrolytes Lost Through Sweat

During activities lasting longer than 50 minutes, substantial amounts of electrolytes such as sodium (salt) and potassium are lost in sweat. These electrolytes are used in muscle contraction, fluid balance, and other body functions, so they must be replaced. In addition, sodium plays a role in activating the body's thirst mechanism, so it can stimulate athletes to keep hydrated. The best way for athletes to replace these nutrients is by drinking a sports beverage (containing sodium) and eating their usual diet. It might be tempting to advise athletes to take salt tablets or other supplements. However, they can replenish sodium and the other necessary electrolytes (like potassium and calcium) through proper nutrition and hydration choices, so using salt tablets is not recommended. Only a small amount of potassium is lost in sweat. Oranges and bananas are good sources of potassium. Having them available during practices and competitions can help athletes replenish the minimal potassium lost through sweat.

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Sports Drinks Versus Water

If athletes are engaged in any vigorous or high-intensity activity, practicing or competing for more than an hour, competing or practicing more than once per day, or are dehydrated, then sports drinks (with 6% to 7% carbohydrate solution and containing sodium) are preferred because they

- stimulate thirst,
- promote fluid retention,
- replenish carbohydrates utilized for energy,
- help to reduce muscle cramping, and
- appeal to athletes (because of the flavor), which causes them to drink more.

Prohibit Artificial Means of Quick Weight Reduction

Athletes looking for the quick and easy way to lose weight may try to cut corners and use unsafe and unhealthy methods to do so. Weight loss should be accomplished by eating healthy, and safely reducing your caloric intake while increasing your caloric expenditure. The 2023-24 NFHS Wrestling Rule Book (Rule 1, Section 5, Articles 1-3) outlines the process for discouraging excessive weight loss and establishing a safe minimum weight, which involves the wrestler, parents or guardians, the appropriate health care provider, and coach (Hopkins 2023).

Identifying and Treating Exertional Heat Illnesses

During physical activity, the body can produce 10 to 20 times the amount of heat that it produces at rest; this heat production occurs through metabolism. Approximately 75 percent of this heat must be eliminated. If the air temperature is less than the body temperature, radiation, conduction, and convection can help dissipate 65 to 75 percent of the heat. However, if the air temperature is near the body temperature, these modes of heat loss are less effective, and the body must rely more on perspiration. High humidity reduces the amount of sweat evaporation, leaving exercising athletes at risk of exertional heat illness.

This section covers three types of exertional heat illness: heat cramps, heat exhaustion, and heatstroke. Each has different signs and symptoms, and each has different first aid interventions. Heatstroke is life-threatening, whereas heat exhaustion and heat cramps typically are not. Therefore, it is important that you learn to evaluate the signs and symptoms and learn to apply the first aid techniques that are appropriate for each illness.

HEAT CRAMPS

Heat cramps are sudden muscle spasms that usually occur in the quadriceps, hamstrings, or calves. Causes include dehydration, electrolyte (sodium and potassium) loss, decreased blood flow to the muscles, or fatigue.

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Signs and Symptoms

- Pain
- Fatigue
- Severe muscle spasms, often in the quadriceps, hamstrings, or calves
- Heavy sweating

FIRST AID

1. Rest the athlete in a cool place.
2. Assist the athlete with stretching the affected muscle.
3. Give the athlete a sports beverage (containing sodium) to drink. If a sports drink is not available, provide the athlete with regular water.
4. If the spasms do not stop with stretching or after a few minutes of rest, look for other possible causes.
5. Monitor the athlete for worsening signs and symptoms, or any signs of dehydration, and send the athlete to the appropriate health care provider if worsening occurs. If spasms continue, or if other injuries are found, inform the athlete's parent or guardian and send the athlete to the appropriate health care provider.
6. Seek medical attention for an athlete who has heart problems, is on a low-sodium diet, or continues to experience cramps for more than an hour.

Playing Status

The athlete can return to activity once the spasms stop and the athlete can run, jump, and cut without limping or experiencing pain. If the athlete is referred to a health care provider, they cannot return to activity until examined and released by the health care provider.

PREVENTION

- Adequate hydration for athletes
- Rest for athletes with signs of dehydration

HEAT EXHAUSTION

Heat exhaustion is a shocklike condition caused by dehydration, which occurs when the body's water and electrolyte supplies are depleted through sweating.

Signs and Symptoms

- Headache
- Nausea
- Dizziness
- Chills
- Fatigue
- Thirst
- Muscle cramps

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- Pale, cool, and clammy skin
- Rapid, weak pulse
- Loss of coordination
- Dilated pupils
- Profuse sweating

FIRST AID

1. Remove the athlete from the hot environment, and rest them from activity and exertion.
2. Remove excess clothing.
3. Send for emergency medical assistance.
4. Have a medical professional take a rectal temperature to differentiate exertional heat exhaustion from the more serious heatstroke. With heat exhaustion, core body temperature (measured rectally) would usually be less than 104 degrees Fahrenheit. Rectal temperature should only be obtained by a trained medical professional. In the event that an athlete has a suspected raised core temperature, full-body cooling should take place immediately and you should send for emergency medical assistance regardless of the actual core temperature.
5. Immediately initiate active cooling by using whole-body cooling (from the neck down). Water should be approximately 33.8 to 78.8 degrees Fahrenheit. If cold-water immersion is unavailable, other cooling methods should be initiated such as commercial ice packs, a cold shower, ice sheets or towels, cooling vests and jackets, or fanning.
6. Give the athlete a sports beverage (containing sodium) to drink. If a sports drink is not available, provide the athlete with water.
7. Monitor breathing, and provide CPR if needed.
8. Monitor and treat for shock as needed.

Playing Status

An athlete must not return to activity on the same day that they experienced heat exhaustion. If the athlete is sent to a health care provider or does not quickly recover, they cannot return to activity until examined and released by the health care provider. The athlete cannot return to activity until the weight lost through sweat is regained.

PREVENTION

- Adequate hydration
- Rest for athletes with signs of dehydration
- Scheduling of activity during cooler times

HEATSTROKE

Heatstroke is a life-threatening condition in which the body temperature rises dangerously high. It is caused by a malfunction in the brain's temperature control center,

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which results from severe dehydration, fever, or inadequate balance of the body's temperature regulation.

Signs and Symptoms

- Feeling extremely hot
- Nausea or vomiting
- Irritability
- Fatigue
- Hot and flushed or red skin
- Lack of sweat
- Very high body temperature
- Rapid pulse
- Rapid breathing
- Constricted pupils
- Diarrhea
- Confusion
- Possible seizures
- Possible unresponsiveness
- Possible respiratory or cardiac arrest

FIRST AID

1. Send for emergency medical personnel.
2. Quickly move the athlete to a cool, shaded area, and rest the athlete from activity and exertion.
3. Place the athlete in a semireclining position. If the athlete is unresponsive, raise their head or roll them on their side as necessary to allow fluids and vomit to drain from the mouth.
4. Remove excess clothing.
5. Have a trained medical professional take a rectal temperature to differentiate exertional heat exhaustion from the more serious heatstroke. With heat exhaustion, core body temperature (measured rectally) would normally be less than 104 degrees Fahrenheit. In the event that an athlete has a suspected raised core temperature, full-body cooling should take place immediately and you should send for emergency medical assistance regardless of the actual core temperature.
6. Immediately initiate active cooling by using full-body cooling (from the neck down). Water should be approximately 33.8 to 78.8 degrees Fahrenheit. If cold-water immersion is unavailable, initiate other cooling methods, such as commercial ice packs, a cold shower, ice sheets or towels, cooling vests and jackets, or fanning.
7. If the athlete is responsive and able to ingest fluids, give them a sports beverage (containing sodium). If a sports drink is not available, provide the athlete with regular water.

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8. Monitor breathing, and provide CPR if needed.
9. Monitor and treat for shock as needed.

Playing Status

The athlete cannot return to activity until examined and released by a health care provider; at that point, they should return to activity gradually.

PREVENTION

- Adequate hydration
- Rest for athletes with signs of dehydration

Cold-Related Illnesses

As with heat-related illnesses, cold-related illnesses are caused by an imbalance in the factors that affect body temperature. Exposure to cold weather and cold equipment causes body temperature to drop below normal. To counteract this drop, the body tries to gain or conserve heat by shivering (increases metabolism) and reducing blood flow to the skin and extremities (to conserve heat and blood flow to the brain, heart, and lungs). This response can result in frostbite or hypothermia.

Prevention of Cold-Related Illnesses

The body best withstands cold temperatures when it is prepared to handle them. Following are guidelines to reduce the risk of cold-related illnesses.

Ensure Athletes Wear Appropriate Protective Clothing

Athletes should dress in layers to allow sweat to evaporate and to protect against the cold. Wool, Gore-Tex, and spandex are excellent materials to wear. Also, be sure the head and neck are covered to prevent excessive heat loss. Mittens are preferable to gloves because they allow the fingers to warm each other.

Keep Athletes Active to Maintain Body Heat

Athletes who must stand along the sidelines should keep moving to help produce body heat. Jumping up and down and jogging in place are good sideline exercises.

Monitor Windchill, and Adjust Cold Exposure Accordingly

The combination of wind, cold temperatures, and wet conditions increases athletes' risk of hypothermia (see figure 11.3 for the U.S. National Weather Service's windchill temperature index). Figure 11.4 is an example of a cold participation policy; it is important to first research your state's individual policies and procedures for cold weather participation.

Monitor Athletes At Risk for Cold-Related Illness

Thin, highly conditioned athletes may be prone to cold-related illness because they have less fat to help insulate their bodies. Dehydrated athletes are also at risk.

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Figure 11.3 *National Weather Service windchill temperature index.*

Air temperature (°F) across columns; wind speed (mph) down rows.

mph	40	35	30	25	20	15	10	5	0	-5	-10	-15	-20	-25	-30	-35	-40	-45
5	36	31	25	19	13	7	1	-5	-11	-16	-22	-28	-34	-40	-46	-52	-57	-63
10	34	27	21	15	9	3	-4	-10	-16	-22	-28	-35	-41	-47	-53	-59	-66	-72
15	32	25	19	13	6	0	-7	-13	-19	-26	-32	-39	-45	-51	-58	-64	-71	-77
20	30	24	17	11	4	-2	-9	-15	-22	-29	-35	-42	-48	-55	-61	-68	-74	-81
25	29	23	16	9	3	-4	-11	-17	-24	-31	-37	-44	-51	-58	-64	-71	-78	-84
30	28	22	15	8	1	-5	-12	-19	-26	-33	-39	-46	-53	-60	-67	-73	-80	-87
35	28	21	14	7	0	-7	-14	-21	-27	-34	-41	-48	-55	-62	-69	-76	-82	-89
40	27	20	13	6	-1	-8	-15	-22	-29	-36	-43	-50	-57	-64	-71	-78	-84	-91
45	26	19	12	5	-2	-9	-16	-23	-30	-37	-44	-51	-58	-65	-72	-79	-86	-93
50	26	19	12	4	-3	-10	-17	-24	-31	-38	-45	-52	-60	-67	-74	-81	-88	-95
55	25	18	11	4	-3	-11	-18	-25	-32	-39	-46	-54	-61	-68	-75	-82	-89	-97
60	25	17	10	3	-4	-11	-19	-26	-33	-40	-48	-55	-62	-69	-76	-84	-91	-98

Frostbite times: 30 minutes; 10 minutes; 5 minutes (see the color categories in the original chart).

Wind Chill (°F) = $35.74 + 0.6215T - 35.75(V^{0.16}) + 0.4275T(V^{0.16})$

Where T = Air Temperature (°F); V = Wind Speed (mph).

Source: National Weather Service. Available: https://www.weather.gov/safety/cold-wind-chill-chart.

Figure 11.4 *Sample cold participation policy.*

Risk	Windchill temperature	Participation guidance
Hypothermia	<50 °F	Proper clothing; No breaks
Hypothermia/frostnip	39–33 °F	Proper clothing; No breaks
Mild frostbite	32–31 °F*	Proper clothing; 15 min indoors after 45 min exposure
Frostbite	30–25 °F*	Proper clothing; 20 min indoors after 30 min exposure
Severe frostbite	25 °F and below*	Indoor activity only

*When windchill is below 31 °F, modify activity and rewarming procedures as necessary.

Adapted from Castellani et al. (2006).

Identifying and Treating Cold-Related Illnesses

Frostbite and hypothermia occur in varying degrees of severity, advancing from mild to severe. Each stage has specific signs and symptoms that will dictate the first aid that you administer.

SAFETY MEASURE

For Frostbite

Do not rub or massage frostbitten areas, apply ice to frostbitten areas, or allow frostbitten tissue to refreeze. Doing any of these things can worsen an athlete's condition.

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FROSTBITE

Frostbite is a condition in which tissues freeze and blood vessels constrict. It is caused by exposure of body parts to cold. There are three stages of frostbite: frostnip (first degree), superficial frostbite (second degree), and deep frostbite (third degree). When experiencing frostnip, the skin will turn red and feel cold to the touch. Superficial frostbite involves localized freezing of the skin and the superficial tissues below it. The nose, ears, toes, and fingers are especially prone to superficial frostbite and may include red or flushed skin that turns white, gray, or pale. Additional signs and symptoms of superficial frostbite include firm, white, and waxy skin; and blisters. The skin may appear tinted purple when the area is rewarmed. Deep frostbite begins superficially, then it advances to deep tissues such as muscles and tendons. Deep frostbite may present with blisters and bluish skin, and the frostbitten area may seem very cold and stiff.

First aid for potential first- or second-degree frostbite includes moving the athlete to a warm area, not allowing the athlete to walk on frostbitten feet, and removing wet and cold clothing. Call the athlete's parent or guardian immediately. Monitor and treat the athlete for shock if necessary, and call for emergency medical assistance if shock occurs. Rewarming should take place only if medical assistance will be delayed or if no risk of refreezing exists. If rewarming is initiated and the area refreezes, it could cause more damage. For potential third-degree frostbite, send for emergency medical assistance immediately, then proceed with guidelines for first- and second-degree frostbite. Monitor breathing, and provide CPR if needed.

An athlete suspected of sustaining a frostbite injury should be released by the appropriate health care provider before returning to activity. To prevent frostbite in athletes, monitor weather conditions during practice and halt practice or change to an indoor location if the weather is cold; ensure athletes wear appropriate clothing during practice; and encourage athletes to stay active at practice.

HYPOTHERMIA

Hypothermia is a condition in which the body temperature drops below 95 degrees Fahrenheit. It is caused by prolonged exposure to a wet, windy, and cold environment; extreme fatigue, such as that experienced after competition in a marathon or triathlon; or dehydration. Mild hypothermia (core temperature of 95 to 90 degrees Fahrenheit) can show the following signs and symptoms: loss of coordination, loss of sensation, drowsiness, shivering, pale and hard skin, numbness, irritability, mild confusion, depression, withdrawn behavior, slow and irregular pulse, slowed breathing, sluggish movements, inability to walk, and difficulty speaking. Moderate hypothermia (core temperature of 89 to 86 degrees Fahrenheit) can show the following signs and symptoms: hallucinations, dilated pupils, decreasing pulse rate, or decreasing respiratory rate. Severe hypothermia (core temperature of 85 degrees Fahrenheit and below) could show unresponsiveness, respiratory arrest, or erratic or no pulse.

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First aid for hypothermia includes checking for responsiveness and normal breathing, sending for emergency medical assistance regardless of severity, and providing CPR if the athlete is not responsive. Move the athlete to a warm area, gently remove cold and wet clothing, and wrap the athlete in blankets. Monitor breathing, and treat for shock as needed.

An athlete with suspected hypothermia cannot return to activity until examined and released by a health care provider. Prevention of hypothermia includes monitoring weather conditions at practices or games, ensuring athletes wear proper clothing during practices or games, and moving practice to a different location if conditions require.

Lightning-Related Injuries

According to the U.S. National Weather Service, individuals in recreation activities and sports comprise two-thirds of all lightning injuries in the United States (NWS n.d.-b). Common sites for these injuries include athletic fields, golf courses, and swimming pools. Metal bats, fences, benches and bleachers, trees, and water are good conductors of lightning. Athletes, staff, and spectators are vulnerable to lightning while around these objects. Because lightning tends to strike taller objects, standing in an open playing field increases the risk of a lightning injury. Seeking shelter under a tree is also risky because lightning can travel through the tree to nearby objects. To minimize the risk of lightning injuries, develop a lightning safety policy.

LIGHTNING INJURY

A direct or indirect hit by lightning can cause a wide variety of injuries, such as burns, fractures, or cardiac arrest. Signs and symptoms of lightning injury could include headache, dizziness, disorientation, burns at the point of entry and exit, or unresponsiveness. First aid should include quickly moving the athlete to a safe area, away from lightning, while also being mindful of any other injuries. Send for emergency medical assistance, monitor breathing, and provide CPR if needed. You should monitor and treat for shock as needed, and remove smoldering clothing, shoes, and accessories such as a belt to prevent burns. If breathing is present and no fractures are suspected, place an unresponsive or incoherent athlete in the recovery position to allow fluids and vomit to drain from the mouth.

An athlete cannot return to activity until examined and released by a health care provider. To help prevent lightning injuries, develop a lightning injury prevention plan, and stop activities when thunder occurs (even without lightning). If you hear thunder or see lightning, move everyone to a safe location. Wait 30 minutes after the last lightning strike or clap of thunder before continuing activity.

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